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RESEARCH

Effect of photobiomodulation on recovery from oral soft tissue local anaesthesia in paediatric patients: a preliminary split-mouth randomized clinical trial

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Abstract

Paediatric dental patients are particularly vulnerable to inadvertent soft-tissue injury following local anaesthesia (LA) This preliminary study evaluated. the clinical efficacy of 810-nm diode laser photobiomodulation (PBM) in reversing oral soft-tissue anaesthesia in paediatric patients. Thirty-four children aged 8–12 years requiring bilateral primary toothextractions under infiltration LA were enrolled in by consideration of ethical codes. Local anaesthesia was administered using 2% lidocaine with 1:80,000 epinephrine, as per the local infiltration anesthesia protocol. Participants were randomly allocated to receive both PBM and sham Laser with 7-10 days washout period between the two interventions. PBM was delivered using an 810-nm diode laser in continuous mode at 200 mW for 12 s per point at six predefined intraoral and extraoral sites. An independent examiner assessed soft-tissue sensation at 5-minute intervals by standardized bilateral palpation, using the non-anaesthetized side as the reference. The primary outcome was time to first assessment at which sensation on the anaesthetized side was judged normal. Data were analysed using a paired-samples t-test. Mean reflection time was significantly shorter in the PBM group (60.7 ± 27.3 min; $p < 0.001$) compared with the Sham lasergroup (78.2 ± 17.0 min). Subgroup analysis demonstrated significant reductions in reflection time in both maxillary and mandibular arches. Photobiomodulation using an 810-nm diode laser demonstrated a reduction in subjective soft tissue anaesthesia recovery time following infiltration LA in paediatric dental patients under the conditions tested.

Keywords Anesthesia recovery · Diode laser · Local anesthesia · Child · Photobiomodulation

Introduction

Pain management is paramount in paediatric dentistry, where local anaesthesia (LA) is widely used to facilitate painless dental care and reduce fear and anxiety. However, prolonged LA action on the soft tissues can cause unfamiliar numbness and inadvertent postoperative behaviours such as eating or speaking, potentially resulting in self-inflicted soft-tissue injuries (SSI). The incidence of SSI has been reported to range from 5% to 18% in children, with higher rates reported in children with special needs [1].

Various strategies have been explored to mitigate lingering soft-tissue anaesthesia (STA) and associated SSI, including Transcutaneous Electronic Nerve Stimulation (TENS), pharmacological methods, and Photobiomodulation (PBM). Passive approaches include removable appliances that mechanically protect the soft tissues until sensation returns,

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as advocated by Alghamidi et al. (2019) [4]. Although such appliances may reduce the risk of SSI, they do not actively reverse anaesthesia and rely on patient compliance [2–4].

TENS has been explored for soft-tissue anaesthesia reversal, although its clinical reliability is questionable [2, 3]. Pharmacological agents such as hydralazine hydrochloride (HCl) and phentolamine mesylate (PM) have been used to accelerate anaesthetic recovery by inducing vasodilation and increasing local blood flow [5]. Despite their effectiveness, these approaches may require additional injections and have been associated with postoperative adverse effects [6–8]. These limitations highlight the need for a safe, minimally invasive, and reliable alternative for Soft Tissue Anaesthesia reversal (STAR) in paediatric patients.

Advances in biophotonics have led to the application of low-power lasers in dentistry. PBM produces non-thermal biological effects and has been associated with changes in adenosine triphosphate (ATP) production, nitric oxide (NO) release, reactive oxygen species (ROS) modulation, gene expression, and anti-inflammatory signalling [9–13]. NO-mediated vasodilation may increase tissue perfusion and facilitate clearance of residual anaesthetic agents, providing a possible mechanism for STA reversal [13].

This study was planned to evaluate the efficacy of PBM using 810-nm diode laser in accelerating the STAR during infiltration local anaesthesia in paediatric patients of age 8–12 years.

Materials & methods

Clinical study design

This single-blinded, randomized, split-mouth clinical trial was approved by the Institutional Review Board and Ethics Committee (IEC***/20**/*******/*****1) and registered with the Clinical Trials Registry of India on 11 April 2023 (CTRI/2023/04/051479||http://ctri.nic.in/). Sample size was determined using G*Power 3.1.9.4 based on the results of Siraj et al. (2021), using a paired-means design with $\alpha = 0.05$, $\beta = 0.01$, and an effect size of 1.7; the analysis indicated a minimum sample size of 34 participants. Children aged 8–12 years visiting the Department of Pedodontics and Preventive Dentistry, ***** ***** *******, **********, requiring bilateral primary tooth extractions under infiltration anaesthesia, were screened for eligibility. Those with systemic conditions contraindicating local anaesthesia, allergies to anaesthetics or recent analgesic use within 24 h, and any injection errors, exceeding the 1 mL volume limit or treatments exceeding 30 min post-anaesthesia administration were excluded. Written informed consent was obtained from parents or legal guardians, and child assent was obtained according to age and understanding.

Subject randomization

A computer-generated block randomization sequence enabled group allocation. In this split-mouth design, one side was randomly assigned to the PBM or sham group at the first visit; at the subsequent visit, the contralateral side received the alternate intervention.

Study procedures.

Lidocaine 2% with 1:80,000 Epinephrine standard 1.8mL cartridge (Lignospan standard, Septodont, USA), 30-gauge needle (Septoject, Septodont, USA) and reusable aspirating syringe (Septodont, USA) were used for LA infiltration of 0.5mL buccal and 0.5mL lingual (total of 1mL) to the teeth planned for extraction. Each participant underwent extractions on one side of their jaw in the first session, followed by the contralateral side 7–10 days later. Both sessions involved extraction under infiltration anaesthesia, following which either laser Photobiomodulation [test group] or sham [control group] was used accordingly.

Photobiomodulation (Test group)

A soft tissue diode laser (810-nm, DenLase CDHC) with biostimulation handpiece and protective eyewear was used. PBM was performed at six predefined sites of them two extraoral and four intraoral sites (**see** Fig. 1**)** [12]. The PBM parameters were calculated accordingly **(see** Table 1**)**. The diode laser device was factory-calibrated, and the output power was verified according to the manufacturer's specifications before commencement of the study.

Sham laser (Control group)

Sham laser application was performed using the same diode laser unit and handpiece, maintaining identical positioning, application sites, duration, and operating sounds as the active PBM procedure. However, the laser emission was disabled internally, ensuring no therapeutic laser energy was delivered to the tissues. The visible guide light was activated to simulate active laser application. Participants wore protective eyewear identical to that used in the test group. This approach ensured effective blinding of only the participants and minimized procedural bias.

Outcome assessments

An independent, trained examiner assessed soft-tissue anaesthesia recovery using a standardized bilateral palpation technique. Assessments were performed every 5 min from the time of LA administration until complete sensory recovery. The anaesthetized side was compared with the corresponding



Fig. 1 Sites of laser photobiomodulation

Table 1 Laser Irradiation parameters for photobiomodulation

Parameter (unit)	Value
Emission mode	Continuous
Power (mW)	200
Beam spot size at target (cm^2)	0.5
Irradiance at target (mW/ cm^2)	400
Exposure duration (s)	12/point
Radiant energy (J)	2.4
Radiant exposure (J/ cm^2)	4.8
Number of points irradiated	6
Area irradiated (cm^2)	3
Radiant exposure/fluence (J/cm2)	4.8
Application technique	Contact mode
Number and frequency of treatment	1
Total radiant energy (J)	14.4

non-anaesthetized side. The examiner categorized the sensation at each assessment as "numb," "tingling," or "normal." "Normal" was defined as a sensation indistinguishable from that of the non-anaesthetized reference side; "tingling" indicated an intermediate or altered sensation; and "numb" indicated reduced or absent sensation. The primary outcome was LA recovery time in minutes, defined as the time from LA administration to the first 5-minute assessment at which the sensation was categorized as normal. Thus, the sensory categories served as clinical descriptors, while the variable entered into statistical analysis was the observed recovery time in minutes. An independent trained examiner other than the operator was considered for evaluation using Bimanual palpation method.

Statistical analysis

The obtained data was tabulated in Microsoft Excel and analyzed using IBM Statistical Package for Social Sciences (SPSS) version 27.0. Results of nominal variables were presented as frequencies and percentages. As quantitative variables were normally distributed between the groups, they were compared using the Paired Samples t-test. A probability value of ≤ 0.05 were considered statistically significant.

Results

Clinical study demographics

Fifty-two children aged 8–12 years requiring bilateral primary tooth extractions under infiltration anaesthesia were screened for eligibility. Following parental consent, 34 children

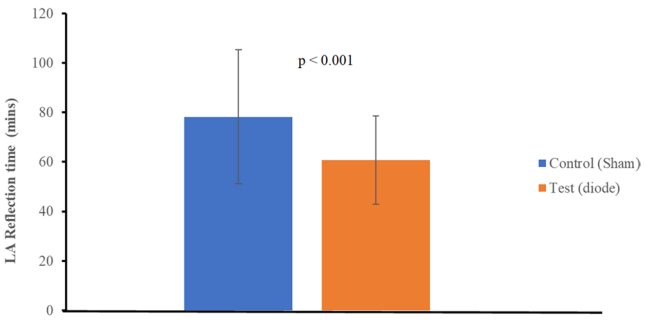


Fig. 2 Comparison of mean LA Reflection time (mins) between groups

Table 2 Intergroup comparison of LA Reflection time (mins)

Group	Mean ± SD	*N*	Mean difference	Std. Error difference	95% confidence interval of the difference		Test value	*p*-value
Control (Sham)	78.20 ± 17.05	34	-17.47	3.05	Lower	Upper	-5.72	< 0.001**
Test (diode)	60.73 ± 27.28	34			-23.68	-11.26		

Paired-samples t-test: $p < 0.01$ – highly significant (**)

exhibiting positive and definite positive behaviour on the Frankel scale who were able to comprehend soft tissue sensation were included. Subjects included females (61.8%) and males (38.2%) with either the mandible (64.7%) or maxilla (35.3%).

LA reflection time

We observed a statistically significant reduction of subjective LA Reflection time in the Test group receiving diode laser Photobiomodulation (60.7 ± 27.3 min) compared to the control group (78.2 ± 17.1 min) with a mean difference of 17.5 min ($p < 0.001$) (**see** Fig. 2; Table 2). This effect was consistent across both mandible (control: 86.3 ± 25.6 min vs. PBM: 67.1 ± 14.5 min, $p = 0.001$) and maxilla (control: 63.3 ± 23.9 min vs. laser: 49 ± 16.2 min, $p = 0.007$) (**see** Fig. 3; Table 3**)**.

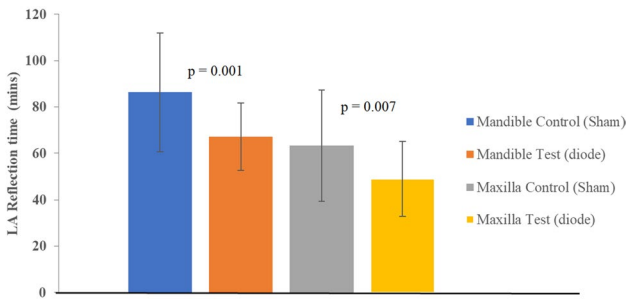


Fig. 3 Comparison of mean LA Reflection time (mins) between groups and Jaw involved

Discussion

Local anaesthesia is essential in pediatric dentistry for pain-free procedures. Yet, its prolonged effect can lead to self-inflicted soft tissue injuries [SSI]. Bagattoni S et *al.* (2020) observed a 15% incidence in children ≤ 6 years and 5% in those > 6 years for SSI [1, 14]. Baillargeau et *al*. (2000) reported higher incidence of SSI accounting to 19% among intellectually disabled children. These findings emphasize the clinical relevance of effective STA management [1, 14, 15]. Various methods have been explored to prevent SSI due to residual STA. Alghamidi WA et *al.* (2019) proposed removable intraoral appliances that protect soft tissues until soft tissue anaesthesia (STA) reversal to prevent SSI. Passive strategies such as removable intraoral appliances offer mechanical protection but do not actively reverse anaesthesia and require patient cooperation, additional fabrication steps, and chairside time, limiting their routine applicability [4].

Active anaesthesia reversal was explored using Transcutaneous Electronic Nerve Stimulation (TENS) by Meechan J G et *al.* (1998) and Hochman R (1988). TENS stimulates rhythmic muscle contractions, enhancing blood circulation and promoting vasodilation and anaesthetic elimination. However, TENS was found to be clinically unreliable [2, 3]. Hersh et *al.* (1995) investigated combinations of 3% Mepivacaine or 4% Prilocaine without a vasoconstrictor instead of 2% lidocaine with epinephrine to accelerate STAR. They, concluded that either of the anaesthetic combinations used did not affect STAR [16]. Wei Zhang et *al.* (2020) synthesized a prodrug (P407-CM-T) that releases tetracaine upon blue light irradiation. This prodrug, a liquid at room temperature and a gel at body temperature, enables

Table 3 Intergroup comparison of LA Reflection time (mins) based on Jaw involved

Jaw involved	Group	N	Mean ± SD	Mean difference	95% Confidence Interval of Difference		Test value	p value
					Lower	Upper		
Mandible	Control (Sham)	22	86.31 ± 25.57	-19.18	-27.63	-10.66	-4.68	0.001**
	Test (diode)	22	67.13 ± 14.49					
Maxilla	Control (Sham)	12	63.33 ± 23.94	-14.33	-23.95	-4.71	-3.28	0.007*
	Test (diode)	12	49 ± 16.23					

Paired-samples t-test: $p < 0.05$ – significant (*); $p < 0.01$ – highly significant (**)

on-demand anaesthesia with tuneable intensity, although it's still experimental. Pharmacological methods for STA reversal include Hydralazine HCl and Phentolamine mesylate [PM]. Fakheran E O et *al*. (2015) found submucosal Hydralazine HCl significantly reduced the time to regain normal sensation (median 81.4 min vs. 221.8 min in the sham group, $p<0.0001$) [5]. There is less clinical evidence supporting it. While PM is well tested and FDA approved method, PM is a non-selective α-adrenergic receptor antagonist, impedes epinephrine's vasoconstrictive action, leading to increased blood flow and faster anaesthetic removal, particularly for anaesthetics co-administered with epinephrine [7, 18]. Varma V et al. (2016) found Phentolamine Mesylate effective in STAR, with significant effect size [Z = -9.04, $P<0.00001$], more pronounced in the maxillary lip (Z = -11.54, $P<0.00001$) than mandibular lip (Z = -3.58, $P=0.0003$)and tongue (Z = -4.69, $P<0.00001$). Sardana et al. (2023) reported PM significantly accelerated sensory recovery in the tongue (effect size: 0.56, 95% CI: 0.41–0.75, $P < 0.001$), lower lip (effect size: 0.56, 95% CI: 0.38–0.83, $P = 0.004$), and upper lip (effect size: 0.50, 95% CI: 0.31–0.80, $P=0.004$) [7]. Recommended PM dose is 0.2–0.4 mg for children aged 4–11 years. Despite its effectiveness, PM requires a second injection and can cause postoperative complications. It's contraindicated in children < 6 years or < 15 kg, and patients with certain cardiovascular conditions. Thus, a safe, non-invasive, effective method is required for STAR [7, 18, 17].

Lasers have revolutionized dentistry, offering precision and minimal invasiveness. Low-level lasers (< 0.1 W/cm²) can modulate tissue and cellular levels, termed as Low-Level laser therapy (LLLT) / Photobiostimulation. / Photobiomodulation[PBM], discovered by Endre Mester (1967), has emerged as a potential therapeutic tool in dentistry for various applications. PBM action is orchestrated through several pathways. Key among these is the interaction with Chromophores and Cytochrome C Oxidase (CCO), a transmembrane macro protein complex in the respiratory electron transport chain. Specific chromophores activate CCO's copper subunit, causing Nitric Oxide (NO) photodissociation, preventing NO's inhibitory effect on cellular respiration. This leads to increased ATP production, Cyclic Adenosine Mono-Phosphate (cAMP) levels and modulated Reactive Oxygen Species (ROS) levels [10, 13]. Another mechanism involves retrograde mitochondrial signalling. PBM is believed to alter the mitochondrial membrane's electrical potential, triggering a signaling pathway that enhances ATP synthesis. PBM also influences cellular concentrations of ROS, Ca2+, and NO, contributing to its therapeutic effects [12, 19]. NO acts on soluble Guanylate Cyclase, elevating cGMP levels. The increased cGMP activates protein kinase G, leading to Ca^{2+} reuptake and opening of calcium-activated potassium channels. The subsequent decrease in Ca^{2+} inhibits Myosin Light-Chain Kinase from phosphorylating Myosin, causing smooth muscle relaxation. Thus, NO production during PBM can enhance vascular perfusion and swift anesthetic removal and STAR [11, 12].

Recent studies suggested PBM for the STAR, which is assumed to be relatively safer than PM [11, 12]. There are no studies that recorded effect the adverse effects of PBM used for STAR hence can be considered for evaluation in future studies. PBM uses red or near-infrared region electromagnetic radiation, with wavelengths usually in the range of 600 to 700 nm and 780 to 1100 nm. Near-infrared light (NIR) (700–904 nm) can be absorbed by the mitochondrial respiratory chain and also, by bio- membrane proteins; whereas red light (600–683 nm) is absorbed by the Semiquinones and Cytochrome Oxidases [19, 20]. Seraj B et al., (2019) advocated the use of 810-nm, while Annu et al.(2023) compared the use of 810 and 660 nm, and reported better reversal of STA with 660 nm [11, 12]. In contradictory to that Pruitt et al. (2023) the 810-nm LED induced measurable and significant rises in blood perfusion [21]. Thus, present study uses 810-nm laser for PBM instead of LED anticipating increased blood perfusion and faster reversal of STA.

Irradiance or power density, recommended for PBM typically falls between 5 mW/cm² and 5 W/cm² [19]. This study employed a power density of 400 mW/cm², which aligns with established recommendations. Earlier researches used a similar power density of 526.31 mW/cm² with a radiant exposure of 6.3 and 6.12 J/cm² to STAR [11, 12]. Notably, our study utilized a lower radiant exposure of 4.8 J/cm². All three studies, including the present one, delivered the radiation in continuous mode. Contact mode is generally recommended for PBM to minimize energy loss [12]. While one of them used non-contact method [11]. The PBM for STAR done at 6 points, 2 points at the injection site, 2 points in the lower lip, 1 point in buccal commissure, and 1 point on the skin below the vermilion at the injection site.

Mark Cronshaw et *al.* (2020) categorized PBM into three groups based on the optical spot sizes employed: small (0.02–0.08 cm²), medium (0.126–0.38 cm²), and large (0.5–4 cm²). The total energy delivered to a single point varied across the groups, with the small spot size group delivering between 0.12 and 18 J, the medium spot size group delivering between 0.3 and 18 J, and the large spot size group delivering between 1.8 and 300 J. The study found a correlation between smaller optical surface applications and clinical success [22]. The present study used 0.5cm² while previous study used 2.28 cm² deliver same energy of 2.4 J, both of them fall under large spot size group.

Among studies those investigated reversal of soft tissue anaesthesia (STA) using PBM the foremost used infiltration anaesthesia (2% lignocaine with 1:100000 epinephrine) while later studies used inferior alveolar nerve block (2% lignocaine with 1:80000 epinephrine) and all the studies were performed on mandible [11, 12, 23]. Phentolamine mesylate and PBM are efficient in reversing local anaesthesia [24]. There is a lacuna in literature regarding the use of PBM for STA reversal in the maxilla, which is addressed by our study. The present study involved children aged 8–12 years (mean 10.0 ± 1.7 years). This age group is older compared to earlier studies [11, 12]. We chose this age group to ensure accuracy, as evaluation of STAR is subjective and younger children may lack the ability to respond reliably, potentially leading to bias. The mean reduction in LA reflection time, in the soft tissues associated with mandibular jaw was 19.18 min. This is less than the 43 min and 55 min reported in previous studies [11, 12]. This may be attributed to sample selection of 8 to 12 years old children. In contrary, prior studies involved younger children (4–8 years) which might be susceptible to detection bias, due to potential limitations in their ability to understand and report the STAR [11, 12, 23]. Apart from age, concentration of epinephrine in LA, method of LA delivery, PBM parameters (irradiance, spot size, method of irradiation) also influence LA reflection time [11, 12, 23, 24, 25]. In case of maxilla mean reduction in LA reflection time was 14.33 min, which is less than the time observed with mandibular infiltration.

Although the reduction in recovery time was statistically significant, its direct clinical significance should be interpreted cautiously. The present study did not evaluate patient-centered outcomes such as incidence of self-inflicted soft tissue injury, postoperative functional discomfort, eating or speaking difficulties, or parental satisfaction. Therefore, whether a reduction of approximately 15–20 min translates into meaningful clinical benefit requires further investigation.

Despite several advancements compared to previous studies, our investigation has certain limitations. First, the assessment of STAR was based on subjective palpation and child-reported sensory responses without the use of validated objective neurosensory assessment tools, which may introduce measurement bias and variability, particularly in paediatric participants. Additionally, although outcome assessment was performed by an independent examiner, complete double blinding was not feasible because the operator delivering PBM was aware of the intervention allocation. Third, only a single PBM parameter protocol was evaluated, without dose–response comparison of different wavelengths, irradiance values, or energy settings, thereby limiting broader generalizability. Future randomized clinical trials with objective outcome measures, inclusion of various anaesthetic techniques, and comparison of multiple laser settings are recommended to address these limitations and strengthen the evidence base.

Conclusions

This preliminary clinical study suggests that 810-nm diode laser PBM may reduce subjective soft-tissue anaesthesia recovery time following primary molar extractions in children aged 8–12 years receiving infiltration LA. However, because the outcome assessment was subjective and patient-centred clinical endpoints were not evaluated, the findings should be interpreted cautiously. Further randomized clinical trials using objective neurosensory assessments, patient-reported outcomes, and comparative PBM protocols are required before definitive clinical recommendations can be made.

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Author contributions A and B had done Conceptualization, Methodology, Supervision, Project Administration, Funding Acquisition, Writing - Original Draft Preparation, Writing - Review & Editing . C, D and E had done Validation, Supervision. F had done Writing - Review & Editing.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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